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CS 499

Module 3-2

8/23/2026

Enhancement Two: Algorithms and Data Structures

For this enhancement, I chose to work on the software design portion in my final project from 360. This was done in April of 2026 and was one of my favorite projects to work on. It is a weight tracking android app developed from the ground up for both the front and back end. This artifact was selected because it has plenty of areas of improvement since I last worked on it. This was my primary goal for this enhancement – to improve the data structure of the project.

The original work had lots of tests that were used to check for failures in some of the features. These would notify the user when the test either passes or fails. In the enhancement, I changed these to only tell the user when they fail and added more tests to newer features that were added. When logging in, the username and password are checked against a comparison to see if that exact username and password are used. When registering, the username and password are registered with capitalizations. If the username already exists, no matter what the capitalizations, the registration fails. If the username and password fields are empty, they also

fail. When deleting a weight, the user is asked if they want to continue. The same is done when the user is deleting their data or user in the settings. They must confirm or deny.